

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **AMARAN**  
 Product Description: **Amaranth**  
 Cat No. : 153030000; 153030250; 153031000; 153035000  
 Synonyms Acid Red 27; C.I. 16185  
 CAS No 915-67-3  
 Molecular Formula C<sub>20</sub> H<sub>11</sub> N<sub>2</sub> Na<sub>3</sub> O<sub>10</sub> S<sub>3</sub>

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company** Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

**Label Elements**
**Hazard Statements**
**Other Hazards**

This product does not contain any known or suspected endocrine disruptors

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

# SAFETY DATA SHEET

Amaranth

Revision Date 21-Mar-2025

| Component                                                                                  | CAS No   | Weight % |
|--------------------------------------------------------------------------------------------|----------|----------|
| 2,7-Naphthalenedisulfonic acid, 3-hydroxy-4-[(4-sulfo-1-naphthalenyl)azo]-, trisodium salt | 915-67-3 | >=85     |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### Self-Protection of the First Aider

No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>), Sulfur oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

# SAFETY DATA SHEET

Amaranth

Revision Date 21-Mar-2025

## Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

## Environmental precautions

Should not be released into the environment.

## Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### **Eye Protection**

Goggles

##### **Hand Protection**

Protective gloves

##### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

# SAFETY DATA SHEET

Amaranth

Revision Date 21-Mar-2025

## Respiratory Protection

No protective equipment is needed under normal use conditions

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                            |                               |                                   |
|------------------------------------------------------------|-------------------------------|-----------------------------------|
| Appearance                                                 | Dark red, Brown               |                                   |
| Physical State                                             | Solid                         |                                   |
| Odor                                                       | Odorless                      |                                   |
| Odor Threshold                                             | No data available             |                                   |
| pH                                                         | 9-10                          | 10 g/L aq.sol                     |
| Melting Point/Range                                        | 218 - 220 °C / 424.4 - 428 °F |                                   |
| Softening Point                                            | No data available             |                                   |
| Boiling Point/Range                                        | No information available      |                                   |
| Flash Point                                                | No information available      | Method - No information available |
| Evaporation Rate                                           | Not applicable                | Solid                             |
| Flammability (solid,gas)                                   | No information available      |                                   |
| Explosion Limits                                           | No data available             |                                   |
| Vapor Pressure                                             | No data available             |                                   |
| Vapor Density                                              | Not applicable                | Solid                             |
| Specific Gravity / Density                                 | No data available             |                                   |
| Bulk Density                                               | No data available             |                                   |
| Water Solubility                                           | 50 g/L (20°C)                 |                                   |
| Solubility in other solvents                               | No information available      |                                   |
| Partition Coefficient (n-octanol/water)                    |                               |                                   |
| Component                                                  | log Pow                       |                                   |
| 2,7-Naphthalenedisulfonic acid,                            | 1.63                          |                                   |
| 3-hydroxy-4-[(4-sulfo-1-naphthalenyl)azo]-, trisodium salt |                               |                                   |
| Autoignition Temperature                                   | No data available             |                                   |
| Decomposition Temperature                                  | > 120°C                       |                                   |
| Viscosity                                                  | Not applicable                | Solid                             |
| Explosive Properties                                       | No information available      |                                   |
| Oxidizing Properties                                       | No information available      |                                   |
| Molecular Formula                                          | C20 H11 N2 Na3 O10 S3         |                                   |
| Molecular Weight                                           | 604.46                        |                                   |

## SECTION 10: STABILITY AND REACTIVITY

# SAFETY DATA SHEET

Amaranth

Revision Date 21-Mar-2025

## Reactivity

None known, based on information available.

## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Excess heat. Avoid dust formation.

## Incompatible Materials

Reducing Agent. Oxidizing agent.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

##### **(a) acute toxicity;**

|                   |                   |
|-------------------|-------------------|
| <b>Oral</b>       | No data available |
| <b>Dermal</b>     | No data available |
| <b>Inhalation</b> | No data available |

### Toxicology data for the components

| Component                                                                                        | LD50 Oral             | LD50 Dermal | LC50 Inhalation |
|--------------------------------------------------------------------------------------------------|-----------------------|-------------|-----------------|
| 2,7-Naphthalenedisulfonic acid,<br>3-hydroxy-4-[(4-sulfo-1-naphthalenyl)azo]-,<br>trisodium salt | LD50 = 6 g/kg ( Rat ) | -           | -               |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

##### **(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

**(e) germ cell mutagenicity;** No data available

# SAFETY DATA SHEET

Amaranth

Revision Date 21-Mar-2025

|                                            |                                                                                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product                                                      |
| (g) reproductive toxicity;                 | No data available                                                                                                                   |
| (h) STOT-single exposure;                  | No data available                                                                                                                   |
| (i) STOT-repeated exposure;                | No data available                                                                                                                   |
| Target Organs                              | No information available.                                                                                                           |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                                                                             |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                                      |
| Symptoms / effects, both acute and delayed | No information available.                                                                                                           |
| Endocrine Disrupting Properties            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

## SECTION 12: ECOLOGICAL INFORMATION

| <u>Ecotoxicity effects</u>                                                                                                                                                                                                                          | Do not empty into drains.                                                                                                                                    |                               |                               |                                                                                            |      |                   |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------|--------------------------------------------------------------------------------------------|------|-------------------|--|--|
| <u>Persistence and degradability</u><br>Persistence                                                                                                                                                                                                 | Persistence is unlikely.                                                                                                                                     |                               |                               |                                                                                            |      |                   |  |  |
| <u>Bioaccumulative potential</u>                                                                                                                                                                                                                    | Bioaccumulation is unlikely                                                                                                                                  |                               |                               |                                                                                            |      |                   |  |  |
| <table><tr><th>Component</th><th>log Pow</th><th>Bioconcentration factor (BCF)</th></tr><tr><td>2,7-Naphthalenedisulfonic acid, 3-hydroxy-4-[(4-sulfo-1-naphthalenyl)azo]-, trisodium salt</td><td>1.63</td><td>No data available</td></tr></table> | Component                                                                                                                                                    | log Pow                       | Bioconcentration factor (BCF) | 2,7-Naphthalenedisulfonic acid, 3-hydroxy-4-[(4-sulfo-1-naphthalenyl)azo]-, trisodium salt | 1.63 | No data available |  |  |
| Component                                                                                                                                                                                                                                           | log Pow                                                                                                                                                      | Bioconcentration factor (BCF) |                               |                                                                                            |      |                   |  |  |
| 2,7-Naphthalenedisulfonic acid, 3-hydroxy-4-[(4-sulfo-1-naphthalenyl)azo]-, trisodium salt                                                                                                                                                          | 1.63                                                                                                                                                         | No data available             |                               |                                                                                            |      |                   |  |  |
| <u>Mobility in soil</u>                                                                                                                                                                                                                             | The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils. |                               |                               |                                                                                            |      |                   |  |  |
| <u>Endocrine Disruptor Information</u>                                                                                                                                                                                                              | This product does not contain any known or suspected endocrine disruptors                                                                                    |                               |                               |                                                                                            |      |                   |  |  |
| <u>Other adverse effects</u>                                                                                                                                                                                                                        | No information available                                                                                                                                     |                               |                               |                                                                                            |      |                   |  |  |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                                                              |                                                                                                                                                                     |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Waste treatment methods</u><br><b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# SAFETY DATA SHEET

Amaranth

Revision Date 21-Mar-2025

|                               |                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point.                                                   |
| <b>Other Information</b>      | Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains |

## SECTION 14: TRANSPORT INFORMATION

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>IMDG/IMO</b>                     | Not regulated                   |
| <b>Road and Rail Transport</b>      | Not regulated                   |
| <b>IATA</b>                         | Not regulated                   |
| <b>Special Precautions for User</b> | No special precautions required |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                                  |            |
|----------------------------------|------------|
| <b>International Inventories</b> | X = listed |
|----------------------------------|------------|

| Component                                                                                  | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------------------------------------------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| 2,7-Naphthalenedisulfonic acid, 3-hydroxy-4-[(4-sulfo-1-naphthalenyl)azo]-, trisodium salt | 213-022-2 | X    | X   | X     | X    | X    | X     | X    | KE-20344 |

### National Regulations

|                                     |                                                                |
|-------------------------------------|----------------------------------------------------------------|
| <b>Persistent Organic Pollutant</b> | This product does not contain any known or suspected substance |
| <b>Ozone Depletion Potential</b>    | This product does not contain any known or suspected substance |

## SECTION 16: OTHER INFORMATION

### Legend

|                                                                                                                              |                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>CAS</b> - Chemical Abstracts Service                                                                                      | <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory  |
| <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances | <b>DSL/NDSL</b> - Canadian Domestic Substances List/Non-Domestic Substances List |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                                                    | <b>ENCS</b> - Japanese Existing and New Chemical Substances                      |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                                             | <b>AICS</b> - Australian Inventory of Chemical Substances                        |
| <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                |
| <b>WEL</b> - Workplace Exposure Limit                                                                                        | <b>TWA</b> - Time Weighted Average                                               |
| <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     | <b>IARC</b> - International Agency for Research on Cancer                        |
| <b>RPE</b> - Respiratory Protective Equipment                                                                                | <b>LD50</b> - Lethal Dose 50%                                                    |
| <b>LC50</b> - Lethal Concentration 50%                                                                                       | <b>EC50</b> - Effective Concentration 50%                                        |
| <b>POW</b> - Partition coefficient Octanol:Water                                                                             |                                                                                  |

# SAFETY DATA SHEET

Amaranth

Revision Date 21-Mar-2025

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date 21-Mar-2025  
Revision Summary Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**